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RESEARCH ARTICLE

A Hybrid Deep Learning Approach for Skin Cancer Classification Using Swin Transformer and Dense Group Shuffle Non-Local Attention Network

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ABSTRACT Skin lesions are localized regions of the skin exhibiting atypical growth or appearance and can be either benign or malignant. Detecting and classifying these lesions accurately is crucial for early diagnosis and effective treatment of skin cancer. However, existing approaches for classifying cancerous and non-cancerous skin lesions from dermoscopic images are limited in capturing the appropriate features and also susceptible to overfitting in imbalanced datasets. This work aims to improve skin cancer classification by enhancing feature representation in dermoscopic images. A hybrid classification system is proposed, combining two tracks: the Swin Transformer and the Dense Group Shuffle Non-Local Attention (DGSNLA) Network, which integrates DenseNet169, Group Shuffle Depth-wise blocks (GSDW), and an enhanced non-local attention block (ENLA). This deep feature fusion approach effectively aggregates global and local features, resulting in more accurate feature representation and improved classification performance. To the best of our knowledge, this is the first study to combine a transformer with a custom CNN for skin cancer classification. The proposed network achieves an accuracy of 94.21%, an F1-score of 96.64%. a precision of 96.62%, and a recall of 96.25% on the HAM10000 dataset.

INDEX TERMS Convolutional neural network, deep learning, dermoscopic images, image processing, skin cancer.

I. INTRODUCTION

Skin lesions are localized abnormalities in the structure of skin which are distinct from the surrounding healthy tissues. These changes can be present from birth, develop later in life, or show up as the first sign of a medical condition. They are mainly of two types, primary and secondary. Primary lesions may be present at birth which can develop later in life due to various factors. They may also arise in response to any disease. Unlike primary skin lesions that appear independently, secondary ones typically arise from transformations of existing lesions. These transformations can be caused by medical interventions, disease progression, or even self-inflicted trauma like scratching [1]. These skin lesions can be cancerous or non-cancerous. One of the most common types of cancer in the world is skin cancer, with an approximated 1.2 million incidences reported in 2020. There is notable increase in melanoma incidence rates among different countries and global regions [2]. Direct exposure to UV radiation results in the long-term damage of skin cell DNA, leading to

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the genetic defects or mutations. These mutations have the potential to induce skin cancer in epidermal keratinocytes and melanocytes [3]. Following exposure to UV radiation, an in-vitro model shows a significant increase in extracellular Lactate Dehydrogenase (LDH) release within 24 hours. This results in early cellular membrane damage. Additionally, there is an increase in IL-1β, inflammatory cytokines, indicating an inflammatory response [4]. Identifying skin cancer at an early stage leads to a 95% increase in the survival rate [5]. Hence, the diagnosis of both benign and malignant melanoma plays a vital role in ensuring efficient and effective treatment for patients.

Physicians utilize various methods to detect skin cancer, with visual inspection being the primary means to identify potential issues. Dermoscopy is a diagnostic imaging technique employed to observe a specific area on the skin, providing the required magnification while mitigating surface reflections. Dermoscopy has proven to be beneficial in the early stages of skin cancer diagnosis, resulting in a significant reduction in fatalities [6]. The lack of objectivity and precision in visually assessing skin lesions leads to extended evaluation times and increased diagnostic errors. Due to the limitations of this approach, the American Academy of Dermatology has introduced a rule known as Asymmetry, Border, Color, Diameter (ABCD) to assist in recognizing the possible characteristics of melanoma which is another technique used [7]. Despite the use of dermoscopy, the professionals may encounter difficulties in distinguishing between various forms of skin cancer. The manual detection often exhibits disparities within classes and slight differences between classes, increasing the likelihood of errors. Hence, integrating Computer-Aided Diagnosis (CAD) is crucial enhancing the efficiency and precision for detection of skin cancer [8].

CAD technology for skin cancer involves four main steps: preprocessing, segmenting lesions, extracting features, and classification [9]. In its early stages, CAD technology faced challenges in accurately diagnosing skin cancer. This is because it relied on existing Machine Learning (ML) methods which had difficulty dealing with complex backgrounds and variations in image contrast. This approach exhibited two limitations. First, the attained accuracy in skin cancer identification did not meet the desired levels, due to the difficulty in differentiating between benign and malignant lesions based solely on handcrafted features. Second, the process of extracting key features from the images proved to be complex and time-consuming, which posed challenges for its implementation in fast-paced clinical settings. In recent years, the emergence of DL has prompted researchers to implement DL-based CAD systems for skin cancer identification. DL automates the identification of complex, non-linear patterns in medical images. This eliminates the manual feature extraction processes required by traditional methods and makes DL more suitable for skin disease recognition.

Despite their strengths, traditional DL methods remain susceptible to certain limitations. These methods are vulnerable to overfitting, particularly in the presence of imbalanced datasets where model focus can shift towards the dominant class [10]. These methods also struggle to effectively capture spatial characteristics and long-range dependencies. This research presents a novel approach with two track network for effective skin cancer detection. The proposed approach uses Swin Transformer in the first track and Dense Group Shuffle Non-Local Attention (DGSNLA) Network in the second track to capture local and global features efficiently.

II. RELATED WORKS

The uncontrolled growth of cells responsible for skin pigmentation can lead to the formation of malignant tumors on the skin. The accurate diagnosis of skin cancer currently relies heavily on the expertise of medical professionals. However, experts can face difficulties in accurately classifying the various types of skin cancer. Recognizing the limitations of manual diagnosis, researchers have used alternative techniques aimed to enhance the accuracy of classification across various skin cancer types through applications of machine learning. This section reviews the existing machine learning and DL models to get better understanding of the current techniques.

Calderón et al. employed a system which concentrates on skin lesion image classification using a DL approach, specifically employing a bilinear CNN architecture. Resnet50, VGG16, and Xception CNNs were utilized to construct two bilinear classification models [1]. Rahman et al. introduced a model employing a weighted average ensemble learning approach. The ensemble makes its predictions on several deep neural network models [6]. Mporas et al. employed a traditional ML approach, incorporating image preprocessing for segmenting images and hair removal. Among the classification algorithms evaluated, the AdaBoost algorithm and random forest classifier achieved better performance [11]. Focusing on the preprocessing of images, Dhivyaa et al. introduced a model that produces high resolution feature maps which are used for extraction of the spatial information of the image. The model used decision trees and random forest algorithms to enhance the reliability of classification of skin images [12]. Similarly, Shetty et al. presented a fully automated system for the skin lesions detection and classification. The model leveraged both machine learning and custom-designed CNN. Inflexibility of traditional machine learning algorithms in handling complex image data and the time-consuming nature of its feature engineering processes paved the way for new technologies to emerge [13].

The recent advancements in DL, particularly its capability for automated feature extraction have facilitated its widespread adoption in the field of medical imaging. A total of 13 DL models were customized, retrained, and individually evaluated for classifying skin lesions was presented by Fraiwan et al. The study analyses the reliability of transfer learning in skin lesion classification [14]. Chaturvedi et al. conducted a comparative analysis on the performance of

five state-of-the-art CNNs. The study presented a deep study to enhance efficacy the proposed ensemble deep learning model for skin lesion classification [15]. Similarly, Jain et al. conducts an analysis of six transfer learning networks for multi-class skin cancer classification. The transfer learning networks involved in the analysis are VGG19, InceptionV3, InceptionResNetV2, ResNet50, Xception, and MobileNet. The findings suggest that Xception Net outperforms the other transfer learning networks employed in the study [16]. Huang et al. built an ensemble model by combining DenseNet and EfficientNet, with EfficientNet-B0 serving as the backbone network. The ensemble achieved its optimal performance through the inclusion of EfficientNet models B1 to B7 (compound scaled networks) [17].

Working on a similar approach, Tajerain et al. developed a system that integrates a EfficientNET-B1, global average pooling 2D layer along with a 7 node softmax layer [18]. Ali et al. constructed a preprocessing image pipeline that involves removing hairs from images, augmenting the dataset, and resizing images according to the specifications of each model within the EfficientNets B0-B7 [19]. Srinivasu et al. introduced a model in which MobileNet V2 was used to classify different types of skin disease. Then LSTM was used to enhance the performance of the model [20]. Salamaa and Aly introduced ResNet50 and VGG-16 models, employing various strategies including different preprocessing methods and the inclusion or exclusion of Support Vector Machine (SVM) [21]. Himel et al. applies the vision transformer architecture for classifying skin cancer. Through the utilization of the self-attention mechanism, the in the image, which allows it to learn relevant features essential for the classifying skin lesions [22]. Mehr et al. introduced a DL-based model that utilized a pretrained Inception-ResNet-v2 network for the image classification. The study also incorporated additional data of patient such as age, gender, and anatomical site [23]. Ramamurthy et al. presented an integrated deep learning model by combining localized, contextual and hierarchical features [35]. This network aimed to improve the feature representation by considering both local, global and hierarchical features for precise classification of dermscopic images. Zhang et al. presented a skin cancer detection approach using recurrent unit networks and orca prediction algorithm [36]. Akilandasowmya et al. presented an ensemble approach for effective classification of skin cancer [37]. The method utilized the Sand Cat Swarm Optimization with ResNet50 (SCSO-ResNet50) to effectively distinguish deep hidden features from known features, thereby enhancing the accuracy of predictions.

Despite the impressive capabilities of DL algorithms, they still are prone to numerous limitations. Their susceptibility to overfitting, particularly in imbalanced datasets, where dominant classes disproportionately influence model learning, leading to poor generalization. Additionally, feature extraction can be limited, which results in inaccurate differentiation between benign and malignant lesions, leading to misdiagnosis.

A. RESEARCH GAPS

The proposed work aims to address the following research gaps in skin cancer classification.

1. Prior research on skin cancer detection primarily focusses on local features and their aggregation. It barely focuses on the information connecting between distant regions, often resulting in suboptimal performance. Hence, this necessitates the development of more sophisticated techniques that can effectively capture both local and global information from skin lesion images.
2. Existing works frequently employ pre-trained models to detect skin cancer, yet they may not consistently differentiate between cancerous and non-cancerous lesions accurately. Consequently, there exists a need to optimize pre-trained models based on specific tasks to achieve precise classification.
3. Although CNN have demonstrated high accuracy in various skin cancer detection and classification tasks, their capacity to capture spatial information and manage non-local correlations is often constrained. This deficiency can be problematic when addressing subtle disease symptoms that necessitate a thorough understanding of the broader context.

B. RESEARCH CONTRIBUTIONS

Research contributions towards the proposed work are as follows.

1. By combining the global and local features extracted from two distinct tracks, the proposed architecture intends to enhance the skin cancer classification. A Swin transformer is used in the first track to extract global features from the input images, and a DGSNLA network is used in the second track to extract local features.
2. By integrating a custom CNN with DenseNet-169, the model leverages the extensive knowledge encoded within the state-of-the-art networks while simultaneously incorporating task-oriented adaptations from the custom network. This approach leads to enhanced performance and faster convergence during the training process.
3. The proposed model leverages GSDW for extraction of spatial variations and fine-grained patterns within the image. It also integrates the ENLA block that captures non-local correlations and models the contextual relationships in the channel enhanced feature map from GSDW.

III. METHODS

The advancement in DL techniques have made significant contributions to medical diagnosis in recent years [24]. In potentially fatal diseases like skin cancer, accurate classification is essential for early diagnosis and timely intervention. This study proposes a model that adopts a dual-track approach that extracts global and local features.

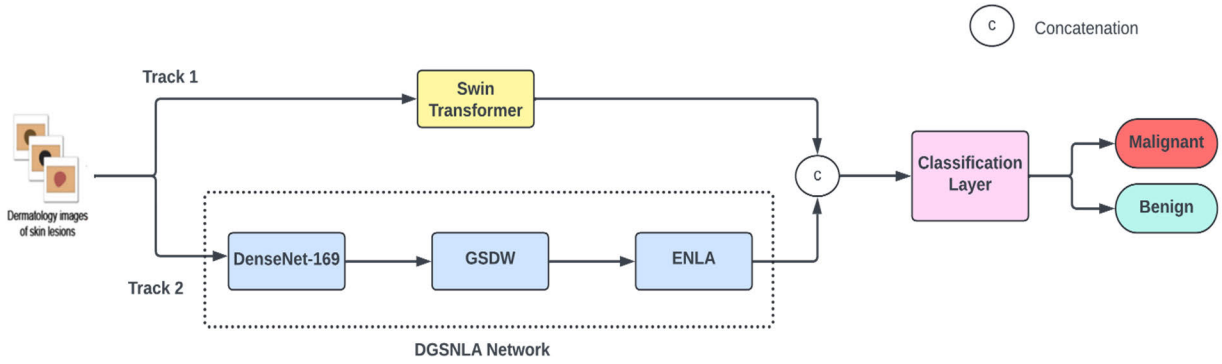


FIGURE 1. Workflow schematic of the proposed network.

Swin Transformer, a state-of-the-art DL network which is used in many image classification tasks, is used in the first track. This network captures long-range dependencies within the image and understands the interaction of different regions of the image. In the second track, incorporating DenseNet-169 as a feature extractor enables the model to capture diverse features effectively. Subsequently, the GSDW block applies depth-wise convolution to each input channel independently, increasing computational efficiency. Subsequently, it incorporates group shuffle operations to facilitate information exchange across the feature maps. This process effectively extracts local features necessary for accurate classification. Then the ENLA block utilizes a non-local attention mechanism, enabling the model to capture long-range dependencies within the input feature map. This mechanism overcomes spatial proximity limitations, and allows the model to incorporate relationships between every pair of positions within the input feature map. Consequently, it effectively models global interactions and enhances feature representation.

The output of the Swin Transformer and DGSNLA network tracks are then fused to get both global and local features. This process effectively combines the complementary information obtained from each track, encompassing both local and global features. By leveraging this fused representation, the proposed model achieves enhanced representational capabilities for skin lesion images, incorporating both spatial and long-range dependencies while capturing the relevant information. The fused features from the both architectures are used for classification of skin cancer. An overview of the proposed network is presented in Fig. 1.

A. SWIN TRANSFORMER TRACK

Swin Transformer is a DL architecture developed for image classification. In this network, the input is partitioned into non-overlapping windows based on a hierarchical approach [25]. Each window is then treated as a distinct token for processing within the network. A key innovation of Swin Transformer is the utilization of a shift window mechanism at each layer. This mechanism facilitates the capture of contextual information between neighboring windows, thereby, enhancing the ability of model to understand the relationships within the image. The architecture of Swin Transformer comprises of multiple stages. Each stage incorporates a combination of attention and convolutional layers. This design enables the model to progressively reduce the spatial resolution of the image, while concurrently preserving fine-grained spatial details crucial for accurate classification. The initial step in Swin Transformer involves dividing the input image into non-overlapping patches with height (H), width (W), and channels numbers. The patches are collectively denoted as P, with a designated patch of size Z. The height of the image patch, denoted as H_p, and width of the image patch, represented as W_p, can be calculated as given in Eq. (1) and Eq. (2), respectively.

$$H_p = \frac{H}{Z} \tag{1}$$

$$W_p = \frac{W}{Z} \tag{2}$$

Swin Transformer blocks differ from traditional architectures by employing Window-based Multi-Head Self-Attention (WMSA) modules instead of standard multi-head self-attention (MSA). Each block processes the previous layer's output Y^t through WMSA, focusing on relationships within localized, shifted windows. The next module calculates Y^{t+1}, which is followed by a two-layer Multilayer Perceptron (MLP) with GELU activation to introduce non-linearity. The final output $\hat{Y}$ of the block is the sum of the WMSA and MLP results, effectively capturing local and global relationships within the image. It can be determined using the Eq. (3).

$$\hat{Y}^t = WMSA\left(L\left(Y^{t-1}\right)\right) + Y^{t-1} \tag{3}$$

where layer normalization is denoted by L and calculated using the Eq. (4).

$$Y^t = MLP\left(L\left(\hat{Y}^t\right)\right) + \hat{Y}^t \tag{4}$$

Following this, the network processes the shifted window. It employs Shifted Window-based Multi-Head Self-Attention

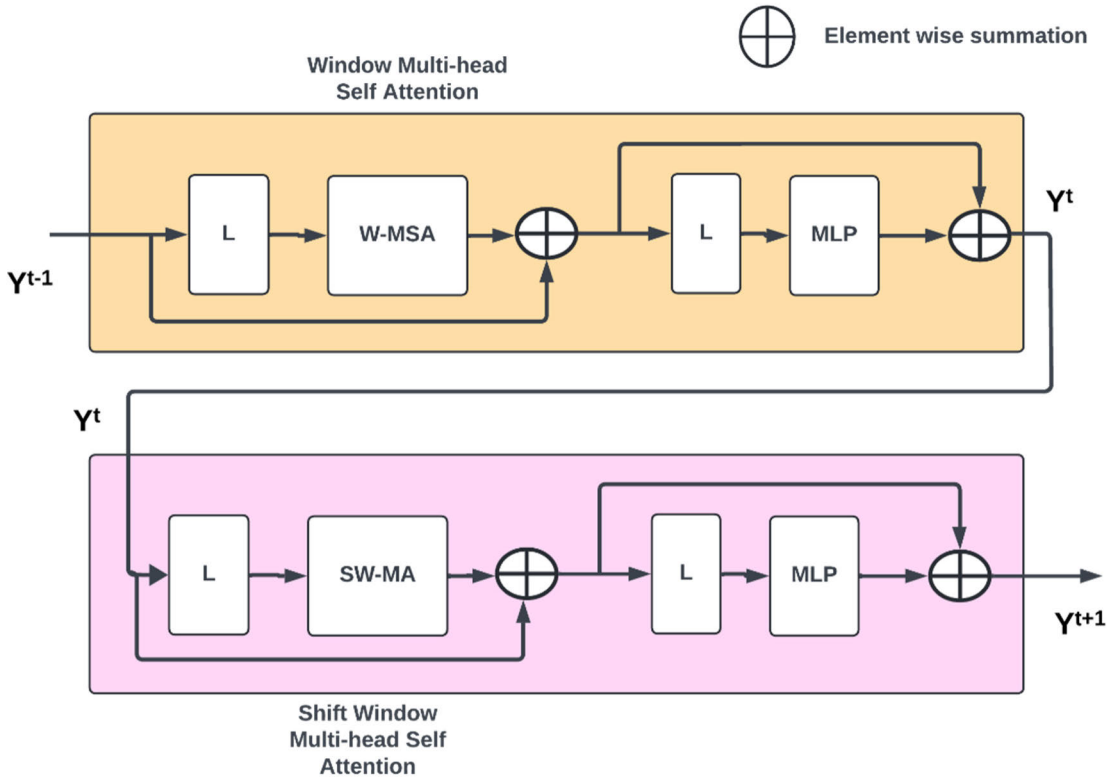


FIGURE 2. Visual representation of the Swin transformer block.

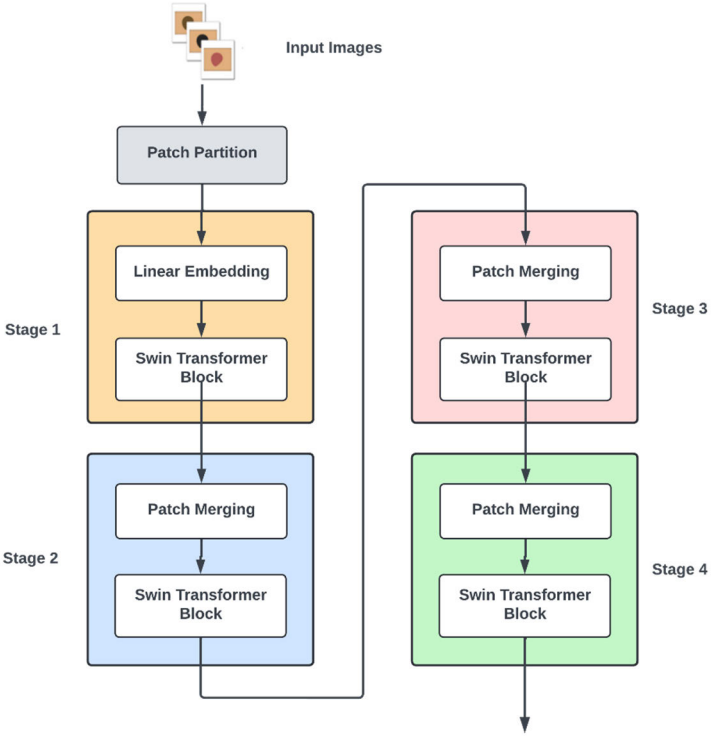


FIGURE 3. Architectural design of the Swin transformer.

(SWMA) on the feature map Y^t, and subsequently uses another MLP module to characterize the feature map. This process results in a new feature map, denoted as $\hat{Y}^{t+1}$ and it is calculated as given in Eq. (5). The formula for calculating

Y^{t+1} is given in Eq. (6)

$$\hat{Y}^{t+1} = SWMA\left(L\left(Y^{t}\right) + Y^{t}\right) \tag{5}$$

$$Y^{t+1} = MLP\left(L\left(\hat{Y}^{t+1}\right)\right) + \hat{Y}^{t+1} \tag{6}$$

The working of the Swin transformer is depicted in Fig. 2 and Fig. 3. The self-attention mechanisms in Swin transformer, unlike standard transformers, operate within local windows while employing a shift window technique. This enables the model to capture global dependencies between pixels across the entire image. This capability to extract long-range dependencies is crucial for accurate object detection and classification tasks.

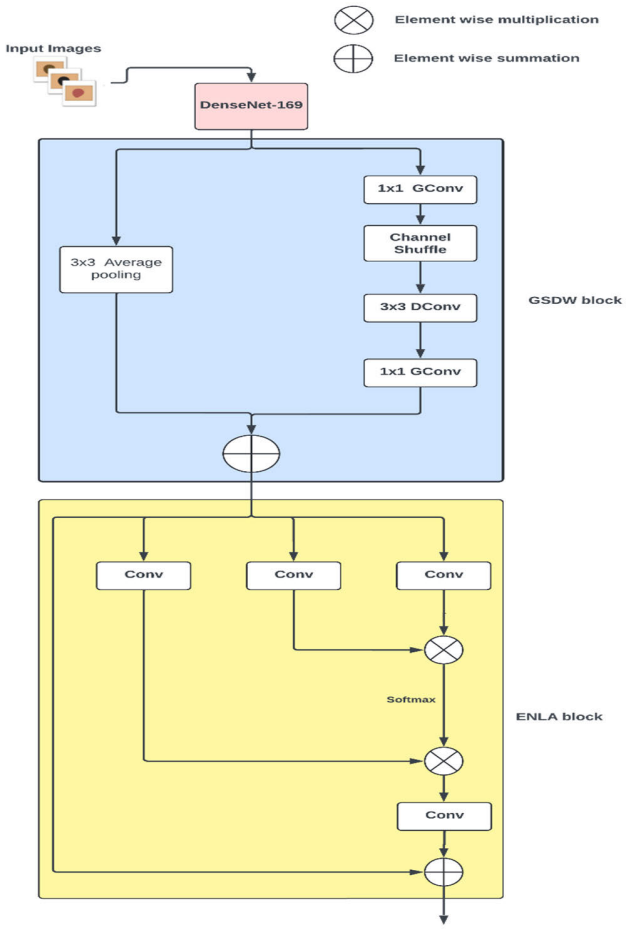


FIGURE 4. Schematic representation of DGSNLA network.

B. DENSE GROUP SHUFFLE NON-LOCAL ATTENTION TRACK

The DGSNLA network track consists of three blocks 1. DenseNet-169 2. GSDW block and 2. ENLA block. Fig. 4 presents the working of the DGSNLA track.

The model integrates a pre-trained DenseNet-169 model as the backbone architecture preceding subsequent blocks. DenseNet-169 is a robust pre-trained network which is trained on a large-scale dataset. There are several advantages of DenseNet-169: it addresses the vanishing-gradient problem, enhances the propagation of features, encourages the reuse of features, and reduces the number of parameters to a significant level. DenseNet-169 is a variant of the DenseNet architecture that specifically comprises 169 layers. It follows the dense connectivity pattern, where each layer receives direct inputs from all preceding layers within its dense block. This facilitates efficient feature reuse and gradient flow throughout the network, enhancing feature representation [26]. DenseNet-169 utilizes dense blocks with convolutional layers, batch normalization, and ReLU activation, along with transition layers to manage feature map dimensions.

By utilizing DenseNet-169 before the subsequent blocks, the model leverages its ability to extract hierarchical features from images efficiently. This is particularly advantageous for tasks requiring capturing intricate patterns and structures. Employing DenseNet-169 facilitates transfer learning, enabling the model to benefit from the learned representations of diverse visual concepts. The features extracted from densenet-169 network are propagated to GSDW block. The key component of the GSDW architecture is the ShuffleNet unit. This unit employs a combination of point-wise group convolutions, depth-wise separable convolutions, and a channel shuffle operation. The GSDW block initially employes group convolution to the input. Three groups are divided from input and each group is distinctly convolved. A 1×1 convolution filter is applied to every channel group within an input feature map during a 1×1 group convolution operation. A 1×1 convolution operati-

on is performed in every group 'i'. The weight kernel 'B' is applied to the relevant input group 'A_i'. The output feature map, denoted as 'Z' and it is calculated in Eq. (7)

$$Z_i = \text{Conv2D}\left(A_i, B_i\right) \tag{7}$$

After channel shuffling, depth-wise convolution is applied within each group. the depth-wise convolution filter is denoted as K which has a kernel size of $K_h \times K_w \times 1$. The convolution operation at position (i, j) in channel C is defined as given in Eq. (8)

$$X_{i,j,c} = \sum_{p=1}^{K_h} \sum_{q=1}^{K_w} X_{(i+p-1),(j+q-1),1} \cdot K_{pqc} \tag{8}$$

Within the output feature map, the value at position (i,j) in channel 'c' is represented by '$X_{i,j,c}$'. This value is derived from the channel-shuffle input tensor, where $X_{(i+p-1),(j+q-1)}$ denotes value at position (i+p-1, j+q-1). Furthermore, $X_{p,q,c}$ signifies the value at position (p,q) within the kernel of depth-wise convolution, applied to the channel 'c.' The channel shuffle operation is the core operation in ShuffleNet. This operation facilitates the flow of information across distinct channel groups within the model. The channel shuffle improves the ability for feature representation of the network and fosters cross-channel communication by allowing each channel to access information extracted from the input

feature map. This shuffling process enables diversification of features and efficient exchange of information between various network segments, enhancing the model's overall performance. Pointwise convolution conducts a 1×1 convolution over all input channels, whereas depth-wise convolution performs independent convolutions on each input channel using separate filters [27]. The GSDW block helps to captures the local features and cross-channel information from the input.

These enhanced feature maps are forwarded to the ENLA block. ENLA is a non-local attention module that capture global information from the input images. This module leverages non-local operations to effectively capture long-range dependencies within the image data [28]. In addition to capturing these non-local correlations, the model also gathers contextual information at the pixel level. This combined approach enhances the model's ability to represent individual pixels within the broader context of the input. For an input feature map x, Eq. (9) defines the non-local operation.

$$y_i = \frac{1}{c(x)} \sum_j f\left(x_i, x_j\right) g\left(x_j\right) \tag{9}$$

Here, let 'i' denote the index representing an output position in the spatial, temporal, or spatio-temporal domain, while 'j' serves as an index enumerating all potential positions. The input signal, denoted as 'x', could represent an image, sequence, video, or features derived from them. The output signal, 'y,' is of equivalent size to 'x.' A pairwise function 'f' calculates a scalar value, representing the relationship, such as affinity, between position 'i' and all positions 'j.' Meanwhile, the input signal at specific location, 'j' is represented by unary function 'g'. A factor C(x) is used to normalize the resultant response. The calculation of C(x) and $f(x_i, x_j)$ is given in Eq. (10) and Eq. (11), respectively.

$$C(x) = \sum_j f\left(x_i, x_j\right) \tag{10}$$

$$f\left(x_i, x_j\right) = \delta\left(x_i\right)^t \theta\left(x_j\right) \tag{11}$$

Here, $\theta(\cdot)$ and $\delta(\cdot)$ represents feature transformations. A residual connection directly links the input feature map to the output of the block. This connection ensures that the original information is preserved and can flow directly to subsequent layers without modification. The DGSNLA network employs GSDW block to extract local dependencies and spatial context within the image. Furthermore, the network incorporates the ENLA block, which is responsible for capturing non-local correlations and modeling the contextual relationships present within the channel-enhanced feature map generated by the GSDW block.

C. CLASSIFICATION

In the classification component, the global features extracted from the Swin Transformer and the local features from the DGSNLA network are concatenated. These concatenated features are then passed through a fully connected layer, which processes the combined features to produce the final output probabilities for each class. This fusion of features, followed by the classification layer, enhances the model's ability to effectively classify skin cancer lesions.

To tackle the class imbalance in the input dataset, the proposed network utilizes the focal loss function, recognized for its effectiveness in enhancing the model's robustness by focusing on classes with fewer samples. This function mitigates class imbalance by assigning lower weights to easier samples while placing greater emphasis on more challenging ones. The network parameters are optimized by maximizing the focal loss of the predicted class probabilities in relation to the target class.

IV. RESULTS AND ANALYSIS

This section presents an in-depth analysis of the research findings. The utilized resources and the environmental conditions under which the experiments were conducted are discussed first. This is followed by a discussion on the strategies employed for hyperparameter tuning, a crucial step in optimizing the proposed network's performance. Furthermore, an ablation study analysis is presented to provide deeper insights into the network's behavior. Finally, the concluding section compares the performance of the proposed network against state-of-the-art models, highlighting the advancements and contributions of this work.

A. DATASET DESCRIPTION

The Human Against Machine with 10,000 training images (HAM10000) dataset is a widely recognized, large collection of dermoscopic images. These images are utilized for the purpose of identifying skin lesions and are openly accessible through the ISIC repository. These images were gathered during a span of 20-year from the skin cancer clinic of Cliff Rosendahl in Queensland, Australia, and the department of dermatology at the Medical University of Vienna, Austria. The dataset consists of a total of 10,015 dermoscopic images categorized into seven distinct classes. These seven classes can be further subdivided into malignant (cancerous) and benign (non-cancerous) groups. More specifically, the benign group comprises 6,705 images belonging to the melanocytic nevi (nv) class, 1,099 to the benign keratosis-like lesions (bkl) class, 142 to the vascular lesions (vasc) class, and 115 images classified under the dermatofibroma (df) class. The malignant group is comprised of 514 images in the basal cell carcinoma (bcc) class, 327 images in the actinic keratoses (akiec) class, and 1,113 images in the melanoma (mel) class [29].

B. DATA AUGMENTATION

The dataset is divided into for training, validation, and testing in 80:10:10 ratio. To tackle the challenge of imbalanced dataset, the data was augmented by horizontal and vertical flipping to each image simultaneously on training and validation sets. These techniques enhance the robustness of the model and concurrently introduce variability while training [30]. Through the combined application of preprocessing and data augmentation techniques, the proposed network achieves accurate classification performance.

C. ENVIRONMENT SETUP AND HYPERPARAMETER TUNING

This study employed an AWS EC2 instance equipped with a 16GB NVIDIA T4 GPU, 4 AMD vCPUs, and 32GB of RAM for image processing and model training. We utilized a ray tuning framework combined with the grid search method to systematically adjust various hyperparameters, including learning rate, epochs, batch size, dropout, and weight decay. Table 1 provides a comprehensive overview of the range of values explored during this optimization process. This procedure significantly impacts the model's accuracy, convergence speed, and generalization capabilities.

TABLE 1. Tuning of hyperparameters for model training.

Parameter	Search Space	Optimal Value
Weight Decay	[0, 1e-3, 1e-4, 1e-5]	1e-3
Learning rate	[1e-3, 1e-4, 1e-5]	1e-4
Batch Size	32 or 64	32
Dropout	[0-0.33]	0.15
Epochs	[30 - 100]	50
Optimizer	[SDG, BGD, ADAM]	ADAM

D. ABLATION STUDIES

The aim of this section is to evaluate the efficacy of major components in the proposed architectural framework to optimize the performance. The efficacy of the following tracks was evaluated: (1) Swin transformer, (2) DGSNLA network, (3) Proposed network. To evaluate the architectural efficiency, a series of experiments were conducted. These experiments assessed the performance of both individual tracks and the proposed network.

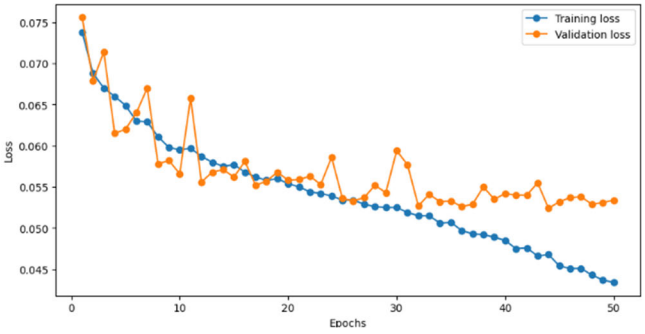


FIGURE 5. Analysis of training phase of the Swin transformer in terms of loss.

1) ANALYSIS OF THE SWIN TRANSFORMER

The performance of Swin transformer is evaluated as a stand-alone component, allowing for a focused examination of its feature extraction capabilities. Along with capturing long-range dependencies, it is also capable of efficiently processing complex images. Fig. 5 and Fig. 6 illustrates the evaluation of training process of the track.

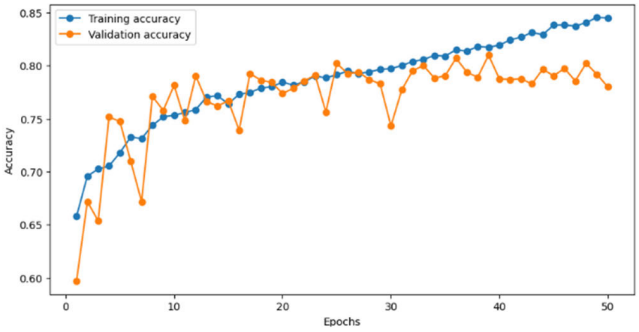


FIGURE 6. Analysis of training phase of the Swin transformer in terms of accuracy.

The model achieved testing accuracy of 85.43%, the average precision, recall, and F1 score were 85.67%, 85.52%, and 85.47%, respectively are demonstrated in Table 2.

TABLE 2. Evaluation metrics for the Swin transformer track.

Class	Precision in %	Recall in %	F1- Score in %
Cancerous	60.15	59.43	59.20
Non-cancerous	93.45	93.38	93.91
Overall	85.67	85.52	85.47

2) ANALYSIS OF THE DGSNLA TRACK

DGSNLA track involves three blocks: DenseNet-169, ENLA and GSDW. This architecture encourages feature reuse and facilitates gradient flow throughout the network, leading to efficient parameter usage and enhanced learning capabilities. GDSW block applies depth-wise convolution to each input feature channel separately for computational efficiency and then introduces group shuffle to encourage information exchange between different parts of the feature maps. This helps in extracting local features from the images. The non-local attention mechanism of ENLA block, allows the model to consider relationships between all pairs of positions in the input feature map, irrespective of their spatial proximity. Fig. 7 and Fig. 8 illustrates performance evaluation of DGSNLA network. The loss plots presented in Fig. 7 and 8 illustrate the training progress up to 30 epochs, at which point convergence had already been achieved. To optimize model performance and prevent overfitting, early stopping criteria were implemented, allowing the training process to halt once the model reached convergence.

The results presented in Table 3 demonstrates an accuracy of 92.32%, F1 score of 93.26%, precision of 93.45%, and recall of 92.88%.

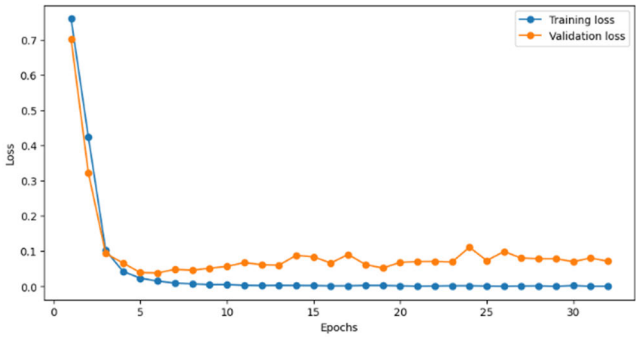


FIGURE 7. Analysis of training phase of the DGSNLA network in terms of loss.

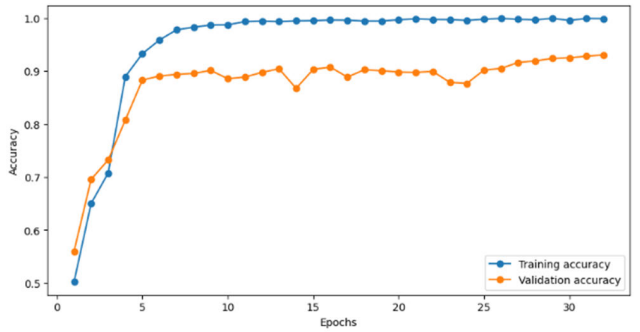


FIGURE 8. Analysis of training phase of the DGSNLA network in terms of accuracy.

TABLE 3. Class-wise evaluation of the DGSNLA network.

Class	Precision in %	Recall in %	F1- Score in %
Cancerous	82.30	82.75	82.61
Non-cancerous	96.98	96.78	97.11
Overall	93. 45	92. 88	93.26

3) ANALYSIS OF THE PROPOSED NETWORK

The combination of DGSNLA track with Swin Transformer was assessed to study the combined effect of both the tracks. The ability of Swin transformer to capture long range dependencies, integrated with the ability of DGSNLA to extract the local features resulted in better classification of skin lesion images. When evaluated against the standalone Swin Transformer and DGSNLA network models, this combined approach achieved superior performance, thus demonstrating the efficacy of their integration. The proposed model leverages both global and local features, enabling it to concurrently capture fine-grained details, inter-dimensional relationships, and long-range dependencies. The model reached optimal convergence after 50 epochs, and the results are visualized in Fig. 9 and Fig. 10.

As Table 4 illustrate, a testing accuracy of 94.21%, along with precision of 96.62%, a recall of 96.25%, and an F1-score of 96.64% was achieved by the proposed network.

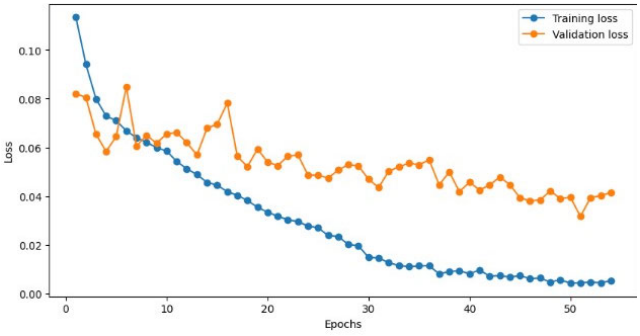


FIGURE 9. Analysis of training phase of the proposed network in terms of loss.

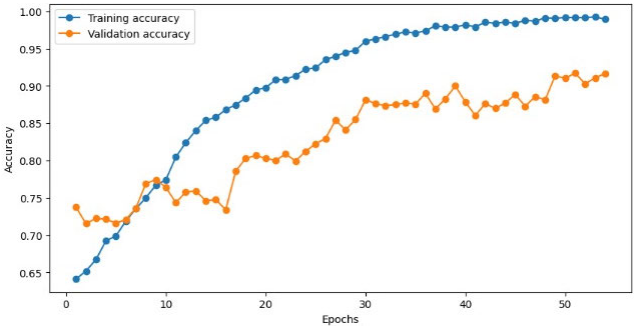


FIGURE 10. Analysis of training phase of the proposed network in terms of accuracy.

TABLE 4. Performance evaluation for the proposed network.

Class	Precision in %	Recall in %	F1- Score in %
Cancerous	87.23	87.51	87.82
Non-cancerous	97.48	97.19	97.39
Overall	96.62	96.64	96.25

The ablation study analyzed the efficiency of proposed model, utilizing a combination of a Swin Transformer and DGSNLA network. The systematic analysis of architectural components in the proposed model demonstrates their significant contribution in enhancing overall performance. It was evident that the Swin transformer, the DGSNLA network, and the proposed model had a significant effect on improving the performance of the architecture. The presented results demonstrate the need of incorporating major components to enhance the efficacy of skin lesion classification. Table 5 summarizes the results of analysis of the performance of various tracks in the proposed network.

E. COMPARATIVE ANALYSIS WITH STATE-OF-THE-ART MODELS

This section evaluates the performance of the proposed model against several established state-of-the-art models, including VGGNet, AlexNet, DenseNet and ResNet. Among the

TABLE 5. Summary of the ablation studies.

Architecture	Accuracy in %
Swin Transformer	85.43
DGSNLA network	92.32
Proposed network	94.21

pre-trained models employed, DenseNet-169 achieved the highest accuracy of 85.7, followed by DenseNet-201 at 85.3%. ResNet-18 exhibited a considerably lower accuracy of 79.6%. As shown in Table 6, the proposed model outperforms the performance of these state-of-the-art models, highlighting its performance in skin cancer classification. This comparative analysis aims to highlight the trade-offs between utilizing state-of-the-art networks and the proposed network specifically developed for the benign and malignant lesion classification tasks.

F. PERFORMANCE COMPARISON WITH EXISTING WORKS

The efficacy of the proposed model is assessed in this section, by comparing it against existing methodologies. Through this comparative analysis between the proposed approach and prior works, we aim to evaluate the strengths and potential enhancements introduced in this study. The evaluation focuses on related studies that have performed binary classification tasks utilizing DL methodologies on the HAM10000 dataset. This comparison includes various ML and DL approaches, including custom CNN and transfer learning. The objective of this comparison is to provide an understanding of the effectiveness of diverse DL models and the application in enhancing the detection of skin cancer. From analysis of Table 7, it was concluded that the deep learning network proposed in the study outperforms the performance of existing DL models with an accuracy of 94.2%. The findings of the study emphasize the potential for customized DL techniques to significantly improve the accuracy of skin cancer classification. This advancement has the potential to contribute meaningfully in the field of dermoscopy diagnosis and ultimately improve the quality of care for patients.

G. LIMITATIONS AND FUTURE WORK

This section focuses on the limitation of the proposed work and the areas for further development and research

1. The proposed approach primarily focuses on the classification of benign and malignant skin lesions. However, it has the potential to be extended to identify specific subclasses of skin cancer. By incorporating additional features and refining the model architecture, future work could enable the differentiation among various skin cancer types, such as basal cell carcinoma, squamous cell carcinoma, and melanoma.
2. To enhance the generalizability and effectiveness of skin cancer detection in real-world applications, future work will focus on incorporating additional features. This includes integrating patient metadata and implementing region-based and gender-based analysis.
3. The scope of this study will be further improved using a broader data range, thus diversifying the sample set.

TABLE 6. Analysis of the proposed network with state-of-the-art networks.

S. No.	Model Trained	Accuracy in %
1	ResNet-18	79.6
2	ResNet-50	82.1
3	VGGNet-16	82.2
4	ResNet-34	83.4
5	ResNet-152	83.7
6	DenseNet-121	83.8
7	AlexNet	84.6
8	DenseNet-201	85.3
9	DenseNet-169	85.7
11	Proposed network	94.2

TABLE 7. Comparison of the performance of the proposed method with existing works.

Source	Method	Accuracy in %
Bechelli et. al. [31]	Custom CNN	82
Ameri A. [34]	Transfer Learning	84
Mehr et. al. [23]	Custom CNN	89.3
Ali et. al. [32]	Transfer Learning	91.3
Salamaa et. al. [21]	Transfer Learning	92
Roge et. al. [33]	Transfer Learning	93.2
Proposed Model	Feature fusion-based classification using Swin transformer and DGSNLA network	94.2

V. CONCLUSION

This study introduces a novel DL framework tailored for classification of cancerous and non-cancerous skin lesions from dermoscopic images. A dual-track feature fusion mechanism is proposed to extract distinctions in skin lesion images. A Swin Transformer is used in first track to extract long-range dependencies inherent in the input data, while concurrently, DGSNLA network in the second track captures spatial and local features. The feature fusion from both tracks facilitates extraction of local and global features. The DGSNLA incorporates a GSDW block that facilitates the efficient extraction of inter-channel dependencies and patterns within the image. Furthermore, the model integrates an ENLA block helps in capturing non-local correlations and modeling the contextual relationships within the channel-enhanced feature map generated by the GSDW block. The model combines a pre-trained DenseNet-169 with a DGSNLA network. By using this approach, the model benefits from the pre-trained knowledge while incorporating adaptations specific to the classification of skin lesion. According to evaluation results on the HAM10000 dataset, the proposed network achieved an accuracy of 94.2%. The findings of this study hold significant implications for dermoscopic image-based diagnostics, offering promising avenues for identification and management of cancerous skin lesions.

DATA AVAILABILITY INTEREST

The HAM10000 dataset is available at: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T.

DECLARATION OF COMPETING INTEREST

The authors declare no competing interest.

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